

6.

Reši enačbo: $AX - XA = I$

$$0 = \text{tr}(AX) - \text{tr}(XA) = n$$

 $\Rightarrow$ protisl.

enačba nima rešitev

$$\text{velja: } \text{tr}(AB) = \text{tr}(BA)$$

7.

 $A \in \text{mat}(m \times n, \mathbb{C})$; vsota $\neq$ vrst. v A a $B \in \text{mat}(n \times p, \mathbb{C})$; — // — B bPokaži, da ima AB konst. vsoto vrstic, izračunaj

$$\sum_{k=1}^n A_{ik} = a \quad \text{za } \forall i \in \{1, 2, \dots, m\}$$

$$\sum_{k=1}^p B_{ik} = b$$

$$(AB)_{ij} = \sum_{k=1}^n A_{ik} B_{kj}$$

$$\sum_{h=1}^p (AB)_{ih} = \sum_{h=1}^p \sum_{k=1}^n A_{ik} B_{kh} = \sum_{k=1}^n a B_{kh} = \underline{\underline{ab}}$$

9. Reši ^{homogenná} [systéma] rovníc:

a) $x + 3y + 2z = 0$

$$x + 5y + z = 0$$

$$3x + 5y + 8z = 0$$

$$\begin{bmatrix} 1 & 3 & 2 & 0 \\ 1 & 5 & 1 & 0 \\ 3 & 5 & 8 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 2 & -1 & 0 \\ 0 & -4 & 2 & 0 \end{bmatrix} \sim \begin{bmatrix} 1 & 3 & 2 & 0 \\ 0 & 1 & -\frac{1}{2} & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$x = -\frac{7}{2}z$$

$$y = \frac{1}{2}z$$

$$\vec{r}(z) = z\left(-\frac{7}{2}, \frac{1}{2}, 1\right)$$

b) $\begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & -1 & 0 & -1 \\ 0 & 1 & 0 & 1 \end{bmatrix} \sim \begin{bmatrix} 1 & 0 & 1 & 0 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$

$$x + z = 0 \rightarrow x = -z$$

$$y + u = 0 \rightarrow y = -u$$

$$\vec{r}(z, u) = (-z, -u, z, u) =$$

$$= \underline{z(-1, 0, 1, 0) + u(0, -1, 0, 1)}$$

$$x, y \in \mathbb{R}^n; A = x \cdot y^T; \text{rk } A = 1$$

$$x^T y = [x_1 \ x_2 \ \dots \ x_n] \begin{bmatrix} y_1 \\ y_2 \\ \vdots \\ y_n \end{bmatrix} = x_1 y_1 + x_2 y_2 + \dots + x_n y_n \leftarrow \text{skalarni produkt}$$

$$A = x y^T = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} [y_1 \ y_2 \ \dots \ y_n] = \begin{bmatrix} x_1 y_1 & x_1 y_2 & \dots & x_1 y_n \\ \vdots & \vdots & \ddots & \vdots \\ x_n y_1 & \dots & \dots & x_n y_n \end{bmatrix} = \begin{bmatrix} x_1 y^T \\ x_2 y^T \\ \vdots \\ x_n y^T \end{bmatrix} \sim \begin{bmatrix} x y^T \\ 0 \\ \vdots \\ 0 \end{bmatrix}$$

$$\Rightarrow \text{rk } A = 1 \checkmark$$

10. Reši.

$$a) \quad x + y + 2z = -1$$

$$2x - y + 2z = -4$$

$$4x + y + 4z = -2$$

$$\begin{bmatrix} 1 & 1 & 2 & -1 \\ 2 & -1 & 2 & -4 \\ 4 & 1 & 4 & -2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & -1 \\ 0 & -3 & -2 & -2 \\ 0 & -3 & -4 & 2 \end{bmatrix} \sim \begin{bmatrix} 1 & 1 & 2 & -1 \\ 0 & -3 & -2 & -2 \\ 0 & 0 & -2 & 4 \end{bmatrix} \sim$$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & -1 \\ 0 & -3 & 0 & -6 \\ 0 & 0 & -2 & 4 \end{array} \right]$$

$$z = -2$$

$$y = 2$$

$$x + 2 - 4 = -1$$

$$x = 1$$

$$\underline{\vec{r} = (1, 2, -2)}$$